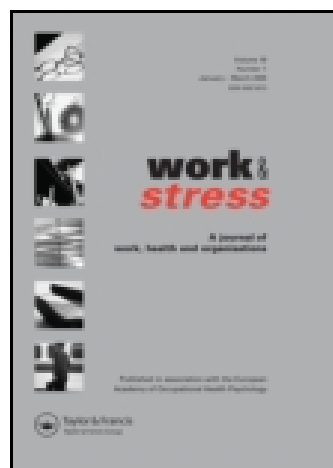


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Outcomes of exposure to workplace bullying: A meta-analytic review

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This paper reports results from two meta-analyses of the potential individual-level outcomes of exposure to workplace bullying. After introducing a theoretical framework for the possible relationships between bullying and outcomes, Study 1 summarizes 137 cross-sectional effect sizes from 66 independent samples ($N = 77,721$). The findings show that exposure to bullying is associated with both job-related and health- and well-being-related outcomes, such as mental and physical health problems, symptoms of post-traumatic stress, burnout, increased intentions to leave, and reduced job satisfaction and organizational commitment. Non-significant or weak associations were established for absenteeism, performance, self-perceptions, and sleep. Study 2 examines longitudinal relationships between bullying and mental health and absenteeism, respectively. Based on prospective associations from 13 samples ($N = 62,916$), workplace bullying influenced mental health problems over time, while baseline mental health problems were associated with a similar increased risk of subsequent reports of exposure to bullying. The long-term effect of exposure to bullying on absenteeism was rather weak. To summarize, the two meta-analyses provide robust evidence for the detrimental effects of workplace bullying that are in line with the theoretical framework presented. The findings have implications for the development of strategies against bullying. Directions for future research are discussed.

Keywords: Meta-analysis; workplace bullying; mental health; outcomes; absenteeism; longitudinal; theoretical model

Introduction

Over the past three decades, a growing body of empirical evidence has established workplace bullying as an important social problem with detrimental implications for those exposed, as well as for organizations and society at large (Einarsen, Hoel, Zapf, & Cooper, 2011a; Tepper & Henle, 2011). Workplace bullying is defined as a situation in which one or several individuals persistently, and over a period of time, perceive themselves as being on the receiving end of negative actions from superiors or coworkers, and where the target of the bullying finds it difficult to defend him or herself against these actions (Einarsen & Skogstad, 1996; Olweus, 1993). That is, while many instances of interpersonal aggression take the form of individual episodes, workplace bullying is by definition characterized by systematic and

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prolonged exposure to repeated negative and aggressive behaviour of a primarily psychological nature, including non-behaviour and acts of social exclusion (Einarsen, Hoel, Zapf, & Cooper, 2011b; Leymann, 1996). As opposed to many other concepts describing interpersonal aggression, workplace bullying is not an either/or phenomenon, but rather a gradually evolving process (Einarsen, 2000). Furthermore, as opposed to, for instance, the concept of abusive supervision, workplace bullying captures aggression from superiors, subordinates and coworkers alike (Tepper, 2007; Zapf & Einarsen, 2011). In line with this, the concept of workplace bullying focuses on the target, as opposed to many other concepts that tend to mainly focus on perpetrators who may behave badly towards many different targets (see also Tepper, 2007). In workplace bullying, it is often the case that a target is singled out and victimized by a range of perpetrators (Zapf & Einarsen, 2011).

Although there is no definitive list of bullying behaviour, bullying mainly involves exposure to verbal hostility, being made the laughing stock of the department, having one's work situation obstructed, or being socially excluded from the peer group. Empirically, such behaviour has been differentiated into seven categories: work-related bullying, social isolation, attacking the private sphere, verbal aggression, the spreading of rumours, physical intimidation, and attacking personal attitudes and values (Zapf, Knorz, & Kulla, 1996). Although individual acts of bullying may be detrimental by themselves, it is the accumulated pattern of behaviour rather than the particular acts that constitute the menace (Leymann, 1990, 1996). Taken individually, such incidents can be seen as only mildly offensive, or at least tolerable, but, cumulatively, they can be experienced as destabilizing, highly distressing and even traumatic to the person at the receiving end.

Theoretically, it is not surprising that workplace bullying is related to the health and well-being of targeted employees. A common feature of most occupational stress models is that stressors in the occupational environment generate negative physical, psychological or behavioural changes in the individual that subsequently manifest themselves in long-term stress reactions such as impaired health, reduced performance and job dissatisfaction (see Jex, 2002 for an overview). The potential negative effects of workplace bullying have been investigated in several studies, and exposure to bullying has been found to be associated with a range of individual-level outcomes, including those mentioned above (see Hoel, Sheehan, Cooper, & Einarsen, 2011; Høgh, Mikkelsen, & Hansen, 2011). However, few exhaustive theoretical models for how bullying is related to different outcome variables have been proposed in the literature to date. In addition, no comprehensive and inclusive systematic statistical synthesis and summary of the whole range of empirical evidence regarding the outcomes of workplace bullying has been conducted so far. In order to remedy these limitations of past research, this paper has a two-fold aim. First, we develop and present a theoretical model that can be used as a framework for both this and future studies on outcomes of bullying. Secondly, we summarize the research literature on a range of individual-level outcomes of exposure to workplace bullying using meta-analysis as the statistical approach. More specifically, by using a Study 1-Study 2 design, we will first investigate reasonable cumulative estimates, given the limitations of the research literature, of cross-sectional associations between exposure to workplace bullying and both job-related and well-being-related outcome variables (Study 1). Since cross-sectional research findings do not provide information about longitudinal relationships and causal associations between variables, the

objective of Study 2 is to statistically summarize longitudinal findings on the individual outcomes of workplace bullying.

To our knowledge, five meta-analyses including studies on workplace bullying have been conducted to date. With regard to the contribution of the present study, it is important to note that two of these studies did not include findings on outcomes at all (Hershcovis et al., 2007; Nielsen, Matthiesen, & Einarsen, 2010), whereas two studies investigated outcomes of aggression by combining findings on abusive supervision, bullying, emotional abuse, generalized abuse, incivility, interpersonal conflict, social undermining and victimization into one overarching first-order phenomenon (Bowling & Beehr, 2006; Hershcovis & Barling, 2010). Hence, following the definition of workplace bullying presented earlier in this introduction (Einarsen & Skogstad, 1996; Olweus, 1993), these two latter studies did not provide nuanced information on workplace bullying as a construct distinct from less intensive and systematic forms of interpersonal aggression (e.g. incivility and social undermining). The only meta-analysis to provide synthesized evidence of the consequences of workplace bullying as a separate and unique phenomenon from other forms of interpersonal aggression was conducted by Hershcovis (2011), whose aim was to compare the outcomes of workplace bullying, incivility, abusive supervision, social undermining and interpersonal conflict, respectively. However, since the Hershcovis study only examined a very limited selection of studies on workplace bullying, and only four outcome variables – job satisfaction (four studies), turnover intent (four studies), psychological well-being (11 studies), and physical well-being (six studies) – its findings could be seen as somewhat preliminary and limited. In addition, it should be noted that all existing meta-analyses have been based on cross-sectional studies only, and have therefore provided no information on the long-term causal associations between workplace bullying and outcomes.

In summary, based on an extensive review of the literature on workplace bullying, the present meta-analysis should make a unique contribution to organizational research by being the first comprehensive and exhaustive statistical synthesis and summary of the empirical evidence regarding the cross-sectional as well as longitudinal individual-level outcomes of exposure to workplace bullying as reported by targets. As workplace bullying has been found to be more strongly associated with well-being-related outcomes than other less intensive forms of interpersonal aggression (Hershcovis, 2011), knowledge about the outcomes of workplace bullying as a distinct phenomenon has important implications for theory, research and practice. In addition to introducing a theoretical model on outcomes of bullying, this study will, by presenting findings on 15 categories of outcome variables, add to theory by providing nuanced information about the relationships between exposure to workplace bullying and a range of important outcomes. With regard to future empirical research, a systematic synthesis of the literature will provide researchers with an overview of what is already known in the field, and thereby provide important guidelines for the direction of future research. As for the practical applications of the study, systematic knowledge about outcomes of workplace bullying has important implications for the development of societal, organizational and individual-level interventions against bullying, including measures aimed at counselling, treating and rehabilitating targets.

Study 1: Cross-sectional evidence on individual outcomes of workplace bullying

With an estimated global prevalence rate for workplace bullying ranging from 11%–18% depending on the measurement method used (Nielsen et al., 2010), and a large body of empirical evidence documenting that exposure to bullying is associated with a range of negative effects for targets, organizations, and society at large (see Hoel et al., 2011; Høgh et al., 2011 for an overview), workplace bullying represents a major problem in contemporary working life. However, as the field is characterized by heterogeneity with regard to measurement methods, samples and populations studied, the external validity of many of the individual studies conducted to date can be questioned (see Nielsen et al., 2010). Although meta-analysis does not resolve the limitations inherent in the existing individual studies, this approach has the advantage of shifting the focus to the whole body of research on a given topic by bringing effects, strengths and limitations of the field into sharper focus. The aim of Study 1 is therefore to summarize cross-sectional evidence in the literature on workplace bullying concerning the relationship between exposure to bullying and outcome variables. We will start by providing a theoretical framework for how exposure to bullying can be related to outcomes, before describing the main categories of individual-level outcome variables explored in the literature. We will then report on the results of a meta-analysis that provides the most comprehensive evidence to date of the magnitude of the direct relationship between exposure to workplace bullying and the outcome variables studied so far.

Outcomes of workplace bullying: A theoretical framework

The scientific study of workplace bullying seems to have arisen from a need to address an important social problem rather than as the result of purely academic or theoretical interest. Theories guiding workplace bullying research are therefore relatively few and far between (Høgh et al., 2011; Zapf & Einarsen, 2005). In response to the lack of a theoretical basis, we suggest a comprehensive theoretical model to explain the individual effects of exposure to workplace bullying. Building on transactional theories of stress (Lazarus & Folkman, 1984), the Affective Events Theory (AET; Weiss & Cropanzano, 1996), and the Cognitive Activation Theory of Stress (CATS; Ursin & Eriksen, 2004), the model presented shows how workplace bullying is associated with individual outcomes. Hence, compared to models on outcomes of aggression in general which mainly have focused on the impact of attributions and role stress (e.g. Bowling & Beehr, 2006; Douglas et al., 2008), this model follows an individual stress-coping perspective. A graphical representation of the model, including an overview of the outcome variables that are to be investigated in this study, is shown in Figure 1.

In line with psychological stress theories in general, the point of departure for our model is that exposure to systematic and prolonged aggression negatively affects the targeted individual (see also Zapf & Einarsen, 2005). The inherent psychological nature of workplace bullying behaviour and the subjective interpretation of being a victim of such treatment (cf. Vie, Glasø, & Einarsen, 2011) indicate that the effects of exposure to bullying can be explained by cognitive mechanisms. Hence, as a way of explaining how these negative effects arise, we suggest a process in which the prolonged duration of a bullying experience develops into repeated and chronic

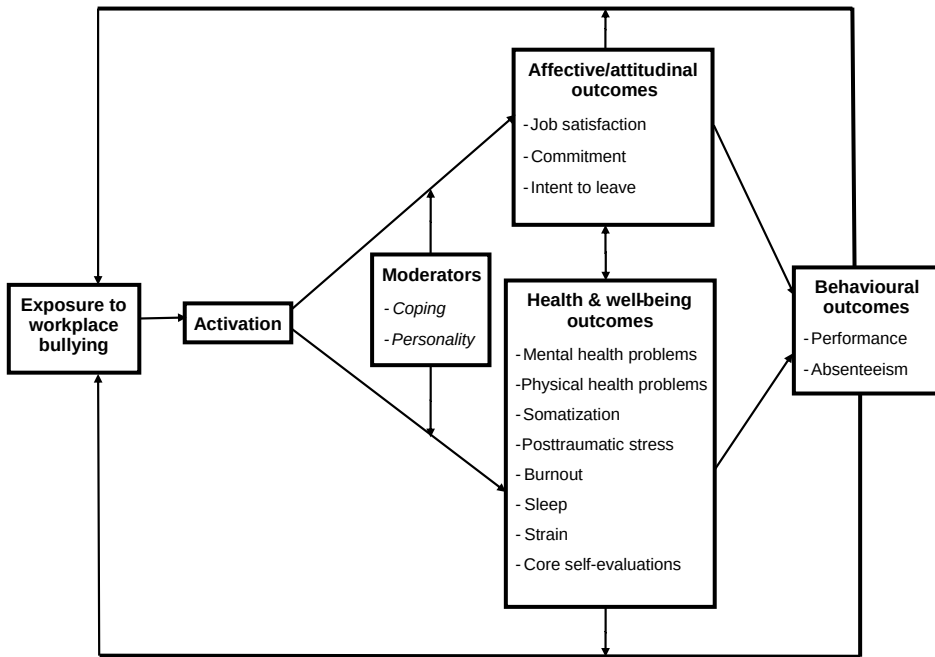


Figure 1. Theoretical model showing the possible relationships between the outcomes of workplace bullying that were included in the meta-analysed studies.

cognitive activation. Building on the core assumptions of CATS, this kind of sustained cognitive activation could, over time, lead to prolonged physiological activation, which is subsequently manifested through impairment of health and well-being (Ursin & Eriksen, 2004).

As empirical evidence has shown that the consequences of bullying vary between individuals, moderating factors must also be taken into consideration. According to transactional stress models (Lazarus & Folkman, 1984), the nature and severity of reactions following exposure to a given stressor are functions of a dynamic interplay between event characteristics and individual appraisal and coping processes. Adopting this perspective, we propose that the severity of outcomes of workplace bullying is dependent upon the interaction between the severity and nature of the bullying behaviour and individual characteristics and coping mechanisms.

Initially, activation following exposure to bullying can be viewed as a normal and healthy response that occurs when there is a discrepancy between what one expects and what actually happens (Eriksen & Ursin, 2004). In cases where the individual sees this discrepancy as manageable and expects to cope fairly well with the situation, activation subsides and does not develop into a health risk. However, when the individual is unable to cope and perceives the discrepancy as unmanageable, the level of cognitive and physical activation may be sustained. An extensive body of research has shown that this kind of cognitive dissonance is associated with persistent high stress levels and pathology, such as decreased sleep quality, increased cortisol levels, elevated heart rate, and increased mortality (Eriksen & Ursin, 2004; Meurs & Perrewe, 2011). In light of this, negative outcomes of exposure to workplace bullying

can be seen as the individual's inability to cope with, and therefore prevent and deter, the unwanted social behaviour of others at work. This idea of failure to cope is further supported by research suggesting that cognitive incongruity between how one perceives oneself and how one is treated by bullies is strongly related to reported health problems among targets of bullying (Mikkelsen & Einarsen, 2002a; Nielsen, Matthiesen, & Einarsen, 2008). Building on the AET, it has also been proposed that the failure to cope with exposure to bullying triggers negative and lasting affective and attitudinal reactions (Glasø, Vie, Holmdal, & Einarsen, 2011). As shown in Figure 1, we further propose that the negative effects of bullying on health, affects and attitudes have secondary and indirect consequences for behavioural outcomes in that reduced health and well-being leads to increased absenteeism and reduced performance. Moreover, as indicated by the feedback loops from the outcome categories pointing to exposure to bullying, it is suggested that potential outcomes of bullying have reversed effects on subsequent exposure to, or at least the reporting of, bullying. This latter association is described in detail in Study 2.

Outcomes of workplace bullying investigated in the literature, and study hypotheses

Recent reviews of the research literature show that workplace bullying has been linked to a range of different outcome variables (Hoel et al., 2011; Høgh et al., 2011). In line with a similar meta-analysis of the consequences of sexual harassment experiences (Willness, Steel, & Lee, 2007), we have chosen to classify these outcomes into two broad categories: job-related outcome variables (attitudes and behaviour) and outcome variables related to health and well-being. With regard to the latter category, the literature has mainly focused on psychological distress reactions such as increased levels of anxiety and depression, and negative core self-evaluations (Mikkelsen & Einarsen, 2002b), and on physical health problems such as musculoskeletal disorders (Høgh et al., 2011). In addition to these acute health reactions, it has also been claimed that workplace bullying is associated with more delayed reactions such as burnout (Einarsen, Matthiesen, & Skogstad, 1998) and post-traumatic stress disorder (PTSD; Leymann & Gustafsson, 1996; Matthiesen & Einarsen, 2004). The former is a disorder characterized by an experience of long-term exhaustion and diminished interest in previously preferred activities, whereas the latter is a diagnosis that refers to a constellation of late stress reactions among people confronted with one or more events that involve a threat to life, serious injury, or that can lead to the death of oneself or others. Although it is debatable whether workplace bullying can be considered to constitute an event that involves life threat, serious injury, or death, as required by the *Diagnostic and Statistical Manual of Mental Disorders* criteria for PTSD (American Psychiatric Association, 2000), it has been argued that the symptoms found among targets of workplace bullying are compatible with those of post-traumatic stress (Leymann & Gustafsson, 1996; Matthiesen & Einarsen, 2004; Tehrani, 2004). Important features of the PTSD condition are persistent re-experiencing of the traumatic experience through memories and dreams, and persistent physical arousal, neither of which was present before the traumatic event (American Psychiatric Association, 2000). Consequently, potential secondary consequences of the mental and physical health problems following exposure to bullying are a heightened strain level and sleep difficulties. In order to meta-analytically summarize findings on the direct relationships between

workplace bullying and outcomes related to health and well-being, the following hypotheses will be examined:

H1: Exposure to workplace bullying is positively associated with mental health problems.

H2: Exposure to workplace bullying is positively related to physical health problems.

H3: Exposure to workplace bullying is positively related to symptoms of post-traumatic stress.

H4: Exposure to workplace bullying is positively associated with symptoms of burnout.

H5: Exposure to workplace bullying is positively associated with sleep problems.

H6: Exposure to workplace bullying is positively related to strain.

H7: Exposure to workplace bullying is negatively associated with core self-evaluations.

Returning to our theoretical model, workplace bullying is also proposed to have implications for job-related outcomes to the extent that targeted employees feel that the organization is partly responsible for the occurrence and frequency of such incidents. Job-related outcomes have therefore been assessed in a range of studies. In what targets may experience as the absence of protective conditions that could help them to cope with the situation, targets can become disillusioned and angry, not only with the bully, but also with the organization itself for failing to protect its employees from such incidents (see Willness et al., 2007). The inability to cope with unwanted and unfair treatment at work can lead to a cognitive dissonance that results in prolonged activation and feelings of discomfort. Being in such a long-lasting uncomfortable state can subsequently develop into dissatisfaction with the job, a lack of organizational commitment and increased intention to leave the job. In order to establish the relationship between workplace bullying and such affective/attitudinal outcomes, we will investigate the following hypotheses:

H8: Exposure to workplace bullying is negatively related to job satisfaction.

H9: Exposure to workplace bullying is negatively associated with organizational commitment.

H10: Exposure to workplace bullying is positively associated with intentions to leave the organization.

Building on theories on revenge and reactions to perceived injustice, workplace bullying is also expected to be associated with absenteeism and productivity (Bies & Tripp, 2005; Moreno-Jimenez, Rodriguez-Munoz, Pastor, Sanz-Vergel, & Garrosa, 2009; Tepper, 2001). If an employee experiences bullying and the bullying is supported and tolerated by the dominant coalition within the work group, the employee may withdraw or disengage as a way of getting even with others in the group and/or the management team. Alternatively, in response to bullying employees may focus on avoiding the bullying or thinking about how to respond. Such a focus could, in turn affect performance.

One may also expect an indirect effect of bullying on absenteeism and performance. A range of studies have established strong relationships between health, well-being, affective variables, and behavioural outcomes. For instance, meta-analytic evidence indicates that job dissatisfaction and lack of commitment are related to lowered productivity and increased absenteeism (Hellman, 1997; Judge, Thoresen, Bono, & Patton, 2001; Scott & Taylor, 1985). Building on these findings, and as illustrated in our theoretical model, it is reasonable to suggest that workplace bullying also has secondary and indirect effects on these kinds of behavioural outcomes. However, because there is a shortage of research on intervening mechanisms in the existing literature on workplace bullying, it is not possible to test the potential mediating effects meta-analytically. Hence, it should therefore be emphasized that this meta-analytic review is not intended to test the causal mechanisms that can explain indirect effects between bullying and behavioural outcomes. Instead, we will assess the direct relationships between the variables in order to shed light on potential associations between workplace bullying and such important outcomes:

H11: Exposure to workplace bullying is negatively related to performance at work.

H12: Exposure to workplace bullying is positively related to levels of absenteeism.

Methodological artefacts as moderators of the bullying-outcomes relationships

The proliferation of ways of measuring and estimating bullying, and extensive use of convenience samples are notable characteristics of research on workplace bullying. Empirical evidence suggests that these methodological issues influence findings on workplace bullying and thereby hamper scientific progress in this field: in a meta-analysis of the impact of methodological moderators of prevalence rates of workplace bullying, it was established that findings on the occurrence of workplace bullying are dependent on moderators such as measurement method (behavioural experiences vs. self-labelling) and sampling procedure (random vs. non-random) (Nielsen et al., 2010). With regard to measurement method, the behavioural experience method means that exposure to workplace bullying is measured using an inventory that includes various types of unwanted and negative behaviour that can be designated bullying if it occurs repeatedly over time. The respondents are then asked to report how frequently they have been exposed to the different behaviours listed in the inventory within a given time period, e.g. during the last six months (Nielsen, Notelaers, & Einarsen, 2011). When the self-labelling method is used, respondents are usually given a single-item question asking whether or not they have been bullied within a specific time period. In some studies, the respondents are offered a theoretical definition of bullying before being asked whether or not they have had experiences in the workplace that correspond to the definition presented (Nielsen et al., 2011). The latter seems to produce a more conservative estimate of workplace bullying than the former (Nielsen et al., 2010). Whereas the self-labelling method measures individual victimization from workplace bullying, the behavioural experience method addresses exposure to a range of bullying behaviours over a period of time. To control for the potential impact of these potential confounding factors in this study, moderator analyses including type of measurement method and sampling procedure were conducted.

Method: Study 1

Literature search

To identify relevant studies, we followed the literature search strategies proposed by Durlak and Lipsey (1991). *Bullying, harassment, mobbing, emotional abuse, and victimization/victimisation* are all concepts that have been used to describe exposure to long-lasting and systematic psychological aggression at work (Einarsen et al., 2011b; Zapf & Einarsen, 2005). They were each combined with the keywords: *outcomes, job satisfaction, intent(ion) to leave, commitment, performance, sick leave, health, depression, psychological distress, burnout, stress, strain, sickness, illness, disease, exhaustion, engagement, anxiety, psychosomatic, somatization, musculoskeletal, and absence* and entered in the PsychINFO, ISI Web of Science, Science Direct, Pubmed, and Proquest databases. Internet searches via www.google.com and Google Scholar were also performed to find other available articles. Papers on related, but less persistent and long-lasting, phenomena such as incivility, social undermining, general workplace abuse, and aggression were screened in order to reveal studies on the phenomenon of workplace bullying being presented under different labels. Further, the authors' personal collection of publications on workplace bullying from around 1988 to the present was examined to find any missing publications. As a final step, citations in the collected publications were inspected. The study coding form was developed by following the guidelines presented by Lipsey and Wilson (2001).

The studies included

To be included in this meta-analysis, studies had to report outcomes of a phenomenon that accorded with the definition of workplace bullying described in the introduction to this paper. Hence, unlike most previous meta-analyses on interpersonal aggression and harassment, which have synthesized many different forms of mistreatment (e.g. Bowling & Beehr, 2006; Hershcovis et al., 2007; Lapierre, Spector, & Leck, 2005), this meta-analysis focuses exclusively on phenomena that fall under the definition of workplace bullying. As a consequence, studies reporting phenomena characterized by one-off incidents or sporadic incidences of aggression and harassment were excluded. Secondly, in addition to theoretically defining the experience of bullying as persistent and long lasting, bullying had to be operationalized either through a measure of self-labelled victimization from bullying or through a behavioural checklist that assessed exposure to bullying behaviour such as the Negative Acts Questionnaire (Einarsen, Hoel, & Notelaers, 2009). Thirdly, studies had to provide the zero-order correlations between one or more of the outcome measures presented in Figure 1 and workplace bullying, or provide sufficient information for these correlations (effect sizes) to be calculated. Studies that lacked this information or reported effect sizes that could not be transformed into correlations were excluded from the analyses.

In line with other meta-analyses within the field (Hershcovis & Barling, 2010), studies that did not include appropriate data (e.g. descriptive data, no measure of bullying or aggression, a higher level of analysis, e.g. group or organizational level, inappropriate variables, theoretical papers, or papers with qualitative data), or for which we were unable to obtain missing information from the corresponding author,

were excluded from the analyses. To avoid double-counting data, the sample should not have been used in a previous study unless different outcome variables were measured. The meta-analytic review was confined to studies published in peer reviewed journals between 1989 (the oldest paper identified) and 2011. The limitation to peer reviewed research papers only was done to assure that the scientific quality of the included studies was satisfactory.

To differentiate between random and non-random samples, we used a criteria and coding scheme developed by Ilies, Hauserman, Schwochau, and Stibal (2003). To be included in the random sample category, an estimate had to be based on (1) a national probability sample (random or stratified random), or (2) a random sample across multiple organizations or in a multiple-site organization (e.g. government or state employees), or (3) a sample resulting from the sampling of the entire sampling frame (as defined by the study) in a single-site organization.

The literature search revealed approximately 500 published papers in scientific journals on a phenomenon that could be classified as workplace bullying, although other terms may have been used. An extensive proportion of these papers were descriptive or review papers that did not contain original data on outcomes of workplace bullying, whereas a range of papers reported on the prevalence of workplace bullying and/or methodological issues. In addition, a large body of studies was restricted to findings on antecedents and causes of workplace bullying such as personality and work environment factors. In total, we identified 54 papers employing a cross-sectional design and 27 papers with a longitudinal design on outcomes of workplace bullying that were in line with our inclusion criteria. In the following, we describe the findings on the cross-sectional studies. The longitudinal papers are further described in Study 2. A list of studies upon which the two meta-analyses are based can be obtained upon request from the first author.

Eight of the studies employing a cross-sectional design provided data from multiple samples, resulting in a total of 66 different samples to be included in the meta-analysis. All studies examined outcomes of workplace bullying from self-reported data provided by targets as opposed to perpetrators or third parties. Most of the included studies reported findings on more than one outcome measure. In cases where a study included multiple measures of the same outcome, the correlation was averaged to create a single study-wide correlation. In all, 137 effect size measures of the relationship between workplace bullying and outcomes were included in this meta-analysis. The total sample size for the 66 samples is 77,721 respondents.

Meta-analytic approach

For all studies, effect sizes were calculated by means of averaged correlations across samples, weighting each observed correlation by the study's sample size. The Q statistic was used to assess the heterogeneity of studies. A significant Q value rejects the null hypothesis of homogeneity. An I^2 statistic was computed as an indicator of heterogeneity in percentages. Increasing values show increasing heterogeneity, with values of 0% indicating no heterogeneity, 50% indicating moderate heterogeneity, and 75% indicating high heterogeneity (Higgins, Thompson, Deeks, & Altman, 2003). As considerable heterogeneity was expected between studies, we calculated the pooled mean effect size using the random effects model. Random effects models are recommended when accumulating data from a series where the effect size is assumed

to vary from one study to the next, and where it is unlikely that studies are functionally equivalent (Borenstein, Hedges, & Rothstein, 2007). Furthermore, random effects models allow statistical inferences to be made regarding a population of studies beyond those included in the meta-analysis (Berkeljon & Baldwin, 2009). The procedure proposed by Hedges and Olkin (1985) was used to investigate potential moderator effects. The presence of a moderator is indicated by a statistically significant Q_B , which suggests a difference between the mean effect sizes across groups.

It is a potential shortcoming of meta-analyses that overall effect sizes can be overestimated due to a publication bias in favour of significant findings. To approach this so-called “file drawer problem” we calculated the fail-safe N which reflects the number of studies reporting null results that would be required to reduce the overall effect to non-significance (Borenstein, Hedges, Higgins, & Rothstein, 2009). All meta-analyses, moderator analyses, and analyses of publication bias were carried out using the Comprehensive Meta-Analysis (version 2) software developed by Biostat (Borenstein, Hedges, Higgins, & Rothstein, 2005).

Results: Study 1

Description of the studies

Seven (13%) of the 54 included studies were published in the 1990s and 36 (67%) in the 2000s, while 11 (20%) studies were published in 2010 or 2011. These numbers reflect the steadily increasing interest in studying workplace bullying. Seventeen (31%) of the studies were from Scandinavian countries (i.e. Denmark, Norway, Sweden), while 23 (43%) studies had other European origin (e.g. Croatia, Finland, Italy, Spain, the UK). Eight (15%) studies originated in North America (USA and Canada), while the remaining studies were from Australia, Japan, and China. With regard to sampling and measurement methods, the majority were based on non-random sampling procedures (66%) and the behavioural experience method (83%). The Negative Acts Questionnaire (Einarsen et al., 2009) was the most frequently used measurement instrument, being employed in 56% of the behavioural experience method studies. The mean response rate in the included samples was 57.8%, which is higher than the average response rate (52.7%) for organizational surveys (Baruch & Holtom, 2008). On average, men and women were equally distributed across the included samples, thus reducing the risk of a potential gender bias. The mean age across the samples was 39 years.

Weighted correlations between workplace bullying and outcomes

Results from the meta-analyses are presented in Tables 1 and 2. With the exception of sleep and core self-evaluations, workplace bullying was significantly associated with all the included health and well-being outcomes (see Table 1). Specifically, workplace bullying was positively associated with symptoms of mental health problems in general ($r = .34$; $p < .001$), as well as its sub-dimensions anxiety ($r = .27$; $p < .001$) and depression ($r = .34$; $p < .001$). Positive associations were also established with regard to symptoms of post-traumatic stress ($r = .37$; $p < .001$), general strain ($r = .31$;

Table 1. Meta-analysis of health and well-being related outcomes of workplace bullying (random effects model).

Outcomes	<i>K</i>	<i>N</i>	Mean <i>r</i>	95% CI <i>r</i>	<i>Q</i>	<i>I</i> ²	NMS
Mental health problems	33	30,785	.34*	.30; .39	439.88*	92.73	1,826*
Anxiety	12	7,863	.27*	.22; .31	45.79*	73.79	1,520*
Depression	17	17,196	.34*	.27; .42	305.21*	94.76	5,187*
Somatization	11	11,733	.28*	.20; .36	178.40*	94.40	2,025*
Physical health problems	11	34,941	.23*	.18; .29	96.37*	89.62	1,227*
Post-traumatic stress	6	1,229	.37*	.30; .43	7.33 ^{ns}	31.81	230*
Burnout	10	4,914	.27*	.19; .35	67.50*	86.67	861*
Core self-evaluations	5	3,941	.07 ^{ns}	-.02; .15	20.52*	80.51	28*
Sleep	4	14,584	-.10 ^{ns}	-.29; .45	729.05*	96.54	0 ^{ns}
Strain	7	6,619	.31*	.19; .41	173.47*	96.54	1,020*

Note: * $p < .001$; ^{ns} = Not significant.

K = number of correlations; *N* = total sample size for all studies combined; *mean r* = average weighted correlation coefficient; 95% CI = lower and upper limits of 95% confidence interval for *mean r*; NMS = Number of missing studies that would bring *p* value to $> \alpha$, based on the fail-safe N method.

$p < .001$), somatization ($r = .28$; $p < .001$), burnout ($r = .27$; $p < .001$), and physical health problems ($r = .23$; $p < .001$).

As shown in Table 2, the relationships between workplace bullying and job-related outcomes were more moderate than the relationships between bullying and health-related outcomes. Significant associations were found, however, with regard to intent to leave ($r = .28$; $p < .001$), job satisfaction ($r = -.22$; $p < .001$), organizational commitment ($r = -.19$; $p < .001$), and absenteeism ($r = .11$; $p < .001$). The association between workplace bullying and performance ($r = -.12$; $p > .05$) was not significant. However, as this latter effect size is based on only three studies, the finding must be interpreted with caution.

Moderator analyses and publication bias

Because of the low number of studies in the outcome categories when sorted by moderator variables, all outcome variables were grouped together in analyses of the moderating impact of measurement method and sampling procedure. All effect sizes were recoded to positive values, and the mean of the included outcomes was calculated for studies that reported more than one outcome variable. The findings

Table 2. Meta-analysis of job related outcomes of workplace bullying (random effects model).

	<i>K</i>	<i>N</i>	Mean <i>r</i>	95% CI <i>r</i>	<i>Q</i>	<i>I</i> ²	NMS
Intent to leave	11	13,205	.28*	.21; .34	127.14*	92.14	2,275*
Commitment	7	3,886	-.19*	-.32; -.05	103.46*	94.2	312*
Job satisfaction	17	17,387	-.22*	-.32; -.12	650.27*	97.54	2,948*
Performance	3	6,385	-.12 ^{ns}	-.26; .02	28.91*	93.08	92*
Absenteeism	9	36,182	.11*	.08; .15	32.82*	75.63	407*

Note: * $p < .001$; ^{ns} = Not significant.

K = number of correlations; *N* = total sample size for all studies combined; *mean r* = average weighted correlation coefficient; 95% CI = lower and upper limits of 95% confidence interval for *mean r*; NMS = Number of missing studies that would bring *p* value to $> \alpha$ based on the fail-safe N method.

from the moderator analyses showed no difference ($Q_B = .04$; $df = 1$; $p = .84$) between random ($r = .29$; $k = 22$) and non-random ($r = .29$; $k = 33$) samples. However, a significant difference ($Q_B = 7.91$; $df = 1$; $p < .01$) in effect size estimates emerged between studies using the behavioural experience method ($r = .31$; $k = 47$) and studies using the self-labelling (with definition) method ($r = .20$; $k = 10$). With the exception of the effect sizes calculated for sleep and core self-evaluations, the fail-safe N estimates indicate that the effect sizes observed in the present meta-analysis are likely to be robust (see Tables 1 and 2).

Discussion and conclusion: Study 1

The findings of Study 1 provide some important information about the potential outcomes of workplace bullying. By synthesizing the literature on the effects of bullying through meta-analysis, the results show that bullying is most strongly related to symptoms of post-traumatic stress and mental health problems. In addition, the findings show that workplace bullying seems to be a major contributor to high levels of intention to leave, and low levels of job satisfaction and organizational commitment. On the other hand, only limited or no associations were established between exposure to bullying and absenteeism, performance, sleep difficulties and core self-evaluations among targets. Overall, the results provide firm support for eight of the 12 proposed hypotheses. The moderator analyses indicate that the sampling procedure does not have an impact on the established effect sizes, whereas a significant effect of the measurement method was established. The publication bias analyses indicate that the robustness of the findings from Study 1 is strong.

The significant associations between exposure to workplace bullying and outcome variables support theoretical assumptions about workplace bullying as a significant social stressor in contemporary working life (Zapf, Knorz et al., 1996). In addition, the associations found between the variables support the theoretical framework guiding this study. The strong relationships between bullying and health-related variables show the importance of developing organizational remedies and measures against workplace bullying, since individual health problems can have severe organizational consequences in a long-term perspective through increased costs related to sick leave, rehabilitation and reduced productivity. That is, although this meta-analysis only established a weak association between bullying and absenteeism and no relationship was found between bullying and productivity, there could be long-term effects that cross-sectional studies are not able to capture. For instance, as most of the studies included in this meta-analysis were based on working populations, it is reasonable to assume that persons on long-term sick leave are excluded from or at least underrepresented in the surveys. In order to investigate the long-term effects of bullying, longitudinal, prospective studies are needed.

With regard to the relatively strong relationship between exposure to bullying and symptoms of post-traumatic stress, it is important to emphasize that this finding is only an indicator of post-traumatic stress-related problems among targets of bullying. In order to use post-traumatic stress disorder as a specific diagnosis, targets have to be screened in a formal diagnostic interview, which was not the case in the studies included in our meta-analysis. Furthermore, because most questionnaires used to assess post-traumatic stress, such as the Impact of Event Scale-Revised (Weiss, 2004), only focus on symptom criteria for post-traumatic stress (DSM-IV

criteria B, C, & D), criteria related to the nature of the traumatic event, symptom duration, and degree of impairment (DSM–IV criteria A, E, & F respectively) are ignored. It therefore remains open to discussion whether post-traumatic stress disorder is an appropriate psychiatric diagnosis in the case of bullying at work (Høgh et al., 2011; Mikkelsen & Einarsen, 2002a).

In addition to providing support for the robustness of this meta-analysis, the findings from the moderator analyses could also have implications for methodological approaches to the study of workplace bullying. First, in line with the conclusions from studies on the assessment of workplace bullying (Nielsen et al., 2010), the difference in effect sizes between the behavioural experience method and the self-labelling (with definition) method suggests that findings concerning outcomes of workplace bullying should be interpreted according to how workplace bullying is measured, and that it is necessary to exercise caution when comparing findings when different methods have been used. Second, the finding that non-random samples provided effect sizes comparable to random samples supports the claim that, although non-random samples cannot be used to provide valid and representative estimates of the occurrence of workplace bullying, they are useful approaches to investigating relationships and phenomenological aspects of workplace bullying (Nielsen & Einarsen, 2008; Nielsen et al., 2010).

There are some methodological considerations that should be taken into consideration in the interpretations of the findings of Study 1. To our knowledge, all the findings on outcomes of workplace bullying reported in the included studies were based on self-report data from self-administered questionnaires or telephone interviews. It should be acknowledged that these kinds of data are most likely influenced by common method bias as well as response set bias such as personal dispositions, mood, expectations, previous experiences, or health. This may cause both non-differential and differential misclassification, resulting in under- and overestimations of effects (Rugulies, 2012). Hence, due to the nature of the included studies, it is not possible to untangle the amount of the aggregate effect size that is a reflection of response artefacts. Secondly, in line with cross-sectional survey research in general, the aggregated effect sizes do not account for the cause and effect relationship between the included variables. To indicate causality, longitudinal study designs are needed.

Finally, it should be noted that the preponderance of European studies in this meta-analysis could be a consequence of the fact that, as a scientific concept, workplace bullying is a relatively new construct that is not yet fully established in many parts of the world, including North America. Related to this, and as discussed in the introduction to this paper, many North American researchers seem to have focused on more overarching constructs that combine various forms of ill treatment and hostile behaviour in the workplace setting, rather than on investigating sub-forms of workplace aggression such as workplace bullying (cf. Fox & Stallworth, 2005; Hershcovis, 2011).

Study 2: Longitudinal evidence of outcomes of workplace bullying

In Study 1, it was established that there are clear relationships between exposure to bullying at work and a range of stress-related outcome variables in targets. However,

because of the cross-sectional design of the studies included, no indications are provided of long-term effects or causal directions between the variables. The aim of Study 2 was therefore to summarize the findings on longitudinal associations between workplace bullying and outcomes using a meta-analytic approach.

Theoretically, both a stressor-strain and a strain-stressor relationship can be expected between workplace bullying and outcomes (Nielsen, Hetland, Matthiesen, & Einarsen, 2012). Going back to Figure 1, the stressor-strain association is shown by the direct effects of bullying on outcomes, whereas the strain-stressor relationship is illustrated by the feedback loops pointing from the various outcome categories to exposure to bullying. In a stressor-strain perspective, it is suggested that workplace bullying generates negative physical, psychological, or behavioural changes in the target that can subsequently result in adverse negative effects among those exposed. In a strain-stressor perspective, individual characteristics such as impaired health and specific personality traits are assumed to constitute a vulnerability factor that increases the risk of being bullied (Bowling & Beehr, 2006). For instance, dissatisfied employees with impaired mental health may elicit aggressive behaviour in others by not living up to expectations, irritating colleagues, and violating social norms for polite social interaction (Einarsen, 2000; Felson, 1992). A strain-stressor relationship can also be explained by the “gloomy perception mechanism” (de Lange, Taris, Kompier, Houtman, & Bongers, 2005). According to this mechanism, unhealthy employees (e.g. depressive workers) report less favourable work characteristics because they evaluate their work environment more negatively than do their colleagues. Hence, employees with mental health problems may perceive their work situation in a more hostile and gloomy way than do their colleagues and therefore report higher levels of workplace bullying (Nielsen et al., 2012). As the stressor-strain and strain-stressor models are not mutually exclusive, the association between bullying and outcomes can also be described as a vicious circle, where bullying is an etiological factor for the outcomes, while outcomes at baseline also predict exposure to bullying (Kivimäki et al., 2003).

In the following, we will investigate longitudinal relationships between workplace bullying and outcome variables by means of meta-analysis. Both the stressor-strain model and the strain-stressor model will be tested. As the existing longitudinal studies on workplace bullying mainly focus on either mental health or absenteeism, this meta-analysis will be limited to these two outcome variables. In addition, strain-stressor relationships will only be examined for the longitudinal relationship between mental health and bullying, since none of the studies identified in the literature search provided information on strain-stressor relationships between absenteeism and bullying. Based on the stressor-strain and strain-stressor models presented above, the following hypotheses will be tested:

H1: Exposure to workplace bullying at baseline is positively related to absenteeism at follow-up.

H2: Exposure to workplace bullying at baseline is positively related to mental health problems at follow-up.

H3: Mental health problems at baseline are positively related to exposure to workplace bullying at follow-up.

Method: Study 2***Literature search and inclusion criteria***

The literature search followed the same procedures as in Study 1, but some additional inclusion criteria were applied: to be included in the meta-analysis, a study needed to report on longitudinal associations between workplace bullying and individual health or absenteeism using at least two measurement points. For the sake of consistency, only studies with a longitudinal design that controlled for the prospective stability of the outcome variable in question were considered for the analysis (see Zapf, Dormann, & Frese, 1996 for a further discussion). Studies that measured bullying at only one point in time and the outcome at another without controlling for the stability of the outcome variable were considered to be mere cross-sectional studies with a time lag. They were therefore excluded from Study 2, but included in Study 1.

Of the total body of peer reviewed articles on workplace bullying, 27 studies employed a longitudinal design. Two studies were excluded in this phase as they did not control for the stability of the outcome variable over time (Moreno-Jimenez et al., 2009; Tepper, 2000). Of the 25 remaining studies, 15 reported findings on either mental health or absenteeism. Two studies were discarded because they lacked effect size indicators (Brousse et al., 2008; Eriksen, Tambs, & Knardahl, 2006), so a total of 13 studies were included in the meta-analysis, each study containing one sample. As most of the studies included were cohort surveys based on relatively large samples, the total sample size for the 13 samples was 62,916 respondents. The same meta-analytic strategy as employed in Study 1 was used to analyse the data. With regard to effect size indicators, odds ratios were calculated in addition to weighted average correlations. Moderator analyses were not conducted because of the low number of studies in each outcome category.

Results: Study 2

Seven of the longitudinal studies included were published between 2000 and 2010, while the remaining six studies were published in 2011. With the exception of one study from the USA (Hoobler, Rospenda, Lemmon, & Rosa, 2010), all the included studies were from Denmark, Finland, Norway and Sweden. While the time lag between baseline and follow-up measurement varied from three to 60 months, the majority of the studies utilized a time lag of between 12 and 24 months.

The results from the meta-analyses of longitudinal effects of workplace bullying on absenteeism and mental health and of the long-term effects of mental health on workplace bullying are presented in Table 3. The findings show that baseline exposure to workplace bullying is significantly related to both absenteeism ($r = .12$; $p < .001$) and mental health problems ($r = .20$; $p < .001$) at follow-up. In addition, a significant reversed effect was found between mental health problems at baseline and increased exposure to bullying at follow-up ($r = .19$; $p < .001$). Analyses of publication bias carried out using the fail-safe N method indicate that these findings are relatively robust.

Discussion and conclusion: Study 2

This meta-analytic study of longitudinal findings on outcomes of workplace bullying replicates and extends the cross-sectional results from Study 1 in that significant

Table 3. Longitudinal associations between bullying and outcomes.

	<i>K</i>	<i>N</i>	Mean <i>r</i>	95% CI <i>r</i>	OR	95% CI OR	<i>Q</i>	<i>I</i> ²	NMS
Bullying – Absenteeism	6	37,054	.12*	.07; .17	1.67*	1.35; 2.07	25.41*	80.32	138
Bullying – Mental health problems	7	28,862	.20*	.14; .26	2.33*	1.76; 3.09	45.94*	86.94	378
Mental health problems – Bullying	4	13,825	.19*	.09; .29	2.02*	1.36; 2.98	15.22*	80.28	75

Note: * $p < .001$.

K = number of correlations; *N* = total sample size for all studies combined; *mean r* = average weighted correlation coefficient; 95% CI *r* = lower and upper limits of 95% confidence interval for correlation coefficient; OR = Odds ratio; 95% CI OR = lower and upper limits of 95% confidence interval for odds ratio; NMS = Number of missing studies that would bring *p* value to $> \alpha$ based on the fail-safe *N* method.

associations are established over time between workplace bullying and the outcome variables of absenteeism and mental health. Moreover, the reversed longitudinal effect of mental health on workplace bullying supports a strain-stressor perspective on the relationship between bullying and outcome variables, and can even point to a circular association between variables. These significant effect sizes thus support all the hypotheses in Study 2, although it should be emphasized that only a weak association was found between bullying and absenteeism. Analyses of potential publication bias indicate that the findings of Study 2 are robust.

The average weighted correlation coefficient of .12 between baseline exposure to bullying and absenteeism at follow-up is in line with the cross-sectional association between the variables established in Study 1. Summarized, this shows that the effects of exposure to bullying on workplace absenteeism were rather weak, but still relatively stable over time. According to Hoel and colleagues (2011), one explanation for the weak effect of bullying on absenteeism is that targets of bullying turn up for work to demonstrate their commitment and loyalty to the organization and to avoid being characterized as malingerers. In some cases, personal guilt in the form of self-inflicted presence could be another factor preventing people from taking time off, even if they, from a medical perspective, would benefit from staying at home. However, building on our theoretical model depicted in Figure 1, in which we propose that the relationship between bullying and absenteeism is explained by the mediating effects of job-related and health- and well-being-related outcomes, an alternative explanation for the rather weak association between exposure to bullying and absenteeism could be that there is an indirect, rather than direct relationship between exposure to bullying and absenteeism. In our meta-analysis, we have only been able to investigate direct effects of bullying on behavioural outcomes, while, theoretically, the relationship is assumed to be indirect. Recent methodological advances have shown that indirect effects between a set of variables can exist even when the direct effects are non-significant (Hayes, 2009). Alternatively, it could simply be that the time lags used in the studies included in the meta-analysis were too short to capture the actual association between bullying and absenteeism, since both bullying and health problems take time to develop. This could also explain why the relationship found between the variables was as strong in the longitudinal as in the cross-sectional studies.

With regard to the association between bullying and mental health, the results show that exposure to bullying behaviour at baseline was significantly related to a subsequent increase in mental health problems, while mental health problems at baseline were associated with an increase in subsequent exposure to workplace bullying. As the stressor-strain and strain-stressor associations between bullying and mental health were almost equally strong, the variables seems to reinforce each other, and the findings support a reciprocal relationship between the variables and thereby the feedback loops from the outcome categories included in our theoretical model presented in Figure 1.

As this meta-analytic summary is based on studies with only two measurement points, one can only speculate about the order of cause and effect between the variables. However, in accordance with the gloomy perception mechanism (de Lange et al., 2005), people with mental health problems might have lower tolerance for exposure to bullying and, consequently, also have a lower threshold for interpreting certain behaviour as bullying (Bowling & Beehr, 2006). Furthermore, in line with a social interactionist perspective on aggression, those with impaired mental health can violate expectations, annoy others, and even violate social norms of polite and friendly interaction (Einarsen, 2000; Felson, 1992) and thus trigger aggressive behaviour in others. This exposure to bullying can in turn increase health problems and thereby develop into a vicious circle.

Some limitations ought to be taken into consideration when interpreting the findings of Study 2. First of all, the reported effect sizes are based on a small number of studies, although they involve rather large and heterogeneous samples. The reversed association between mental health and bullying, for instance, is calculated on the basis of only four studies, all of which originate in a specific geographical region in northern Europe. Another limitation is that the studies included employed different time lags between baseline and follow-up measurements, although most of them utilized a lag of between 12 and 24 months. Hence, it is plausible that reported effect sizes could vary between different time lags. Because of the small number of studies included and the somewhat biased study characteristics, more research is needed on longitudinal associations between bullying and outcome variables in order to increase our knowledge about how workplace bullying influences targets.

General discussion and conclusion

The main aim of this Study1-Study 2 meta-analytic summary of the academic literature on workplace bullying was to increase our knowledge about how exposure to workplace bullying is related to potential outcome variables. In Study 1, we have shown that bullying is associated with a range of different outcome variables, but especially with variables relating to the mental health and well-being of those exposed to it. In Study 2, we have established that these associations persist over time, revealing both a stressor-strain and a strain-stressor relationship between bullying and outcomes. In light of our two quantitative reviews, we will summarize our findings in the final part of this paper and make suggestions for future research that could further the understanding of the nature of the relationship between workplace bullying and individual-level stress-related outcomes.

In reading this general discussion it should be kept in mind that our two meta-analyses are based exclusively on published studies; unpublished studies were excluded. Although it has been suggested that researchers should aim at including unpublished literature in meta-analyses and systematic reviews, our standpoint is that the inclusion of data from unpublished studies can itself introduce bias (Higgins & Green, 2011): First of all, the studies that can be located is likely to yield an unrepresentative sample of all unpublished studies. For instance, the identification of unpublished studies may depend on the willingness of investigators of unpublished studies to provide data. This may depend upon the findings of the study, with more favourable results being provided more readily (Higgins & Green, 2011). Secondly, unpublished studies may be of lower methodological quality than published studies. In a study of 60 meta-analyses that included published and unpublished trials it was found that unpublished trials were less likely to conceal intervention allocation adequately and to blind outcome assessments (Egger, Juni, Bartlett, Hoenstein, & Sterne, 2003). Hence, as our meta-analyses are based on a comprehensive literature search of studies published in peer reviewed journals, the scientific quality of the included studies should be ensured while they also should be representative of the published literature on workplace bullying.

As shown in the two meta-analyses, the published literature on workplace bullying is characterized by a large body of studies focusing on outcomes such as mental health, physical health and job satisfaction, whereas there is a clear shortage of studies on other outcome variables. There is also a lack of full-panel longitudinal studies. Our first request for future research is therefore that upcoming studies should include these more rarely investigated outcome variables rather than focusing solely on health-related outcomes, and preferably use prospective study designs. Based on our summary of the literature and its research findings, we have provided strong evidence for the conclusion that exposure to bullying is related to mental health outcomes both cross-sectionally and longitudinally. However, with regard to theoretical and empirical advances in the field, more research is needed to establish the relationship between exposure to bullying and possible outcome variables such as sleep and productivity.

Having established a relationship between workplace bullying and outcomes such as mental health, intentions to leave and job satisfaction, the next logical step is to investigate mechanisms that can explain how and when the variables are related. So far, only a few studies have examined such mechanisms (e.g. McCormack, Casimir, Djurkovic, & Yang, 2009; Mikkelsen & Einarsen, 2002a; Nielsen et al., 2008). Hence, there is a need for both theoretical explanations and empirical assessments of moderating and mediating variables that can explain the effects of bullying. With regard to theoretical explanations, we have proposed a comprehensive theoretical model, including both moderating and mediating variables, that can serve as a fruitful starting point and we will therefore proceed to test this model in the study of stress-related outcomes of workplace bullying. However, since research on bullying lacks a clear theoretical basis, we encourage all attempts at further theory development.

As cross-sectional studies have limited value with regard to understanding cause and effect relationships between variables, future research should use longitudinal designs, preferably with three or more measurements points (cf. Ployhart & Vandenberg, 2010). With regard to the time lag between survey points, both short

and long lags have the potential to provide valid knowledge about the relationship between bullying and outcomes. However, since bullying is a dynamic process that takes place over a prolonged time period, large-scale diary studies with frequent measurement points could also be a valuable and appropriate approach. In addition, as most surveys of workplace bullying are based on self-reported data, there is a high risk of biases in the data, for example, common method variance and response set tendencies. If possible, future studies in the field should include more objective outcome data, such as registered sick leave and objective or at least third-party performance ratings in order to reduce the risk of self-report bias. As regards health outcomes, some studies of bio-markers such as cortisol do exist (Hansen et al., 2006; Hansen, Høgh, & Persson, 2011), but more such studies are definitely needed (Høgh et al., 2011). Aggregating individual ratings of variables such as bullying and productivity to the group and organizational level could also provide more objective evaluations of outcomes.

Having shown that bullying is associated with a range of negative outcome variables, our final suggestion is to increase the efforts to understand how workplace bullying can be prevented and handled. In our review of the literature in connection with this meta-analysis, we noticed a substantial lack of studies of preventive measures against bullying. By illuminating the consequences of workplace bullying, this meta-analysis will hopefully stimulate additional research aimed at understanding, and ultimately preventing, workplace bullying.

Note

A list of studies upon which the two meta-analyses are based can be obtained on request from the first author.

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